

AYUSH RAJ

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EXPERIENCE

Software Developer Intern ↗

Horizon Mars Rover Team, CUSAT

📅 Nov 2023–Present 📍 Kochi, Kerala

- Developed and implemented a Unified Robot Description Format (URDF) -based robotic arm model with Arduino Due (32 bit ARM Micro-controller) and servo motor control in Robot Operating System (ROS).
- Technologies used : C++, ROS 2 (Humble), Linux, Arduino IDE

AI - ML Intern ↗

IBM Skillsbuild

📅 June 2024 – July 2024 📍 Remote

- Developed expertise in neural networks, NLP, and GenAI, built a kidney stone detector and a fine tuned chatbot which was integrated into a restaurant web portal with 20% faster menu search.
- Technologies used : IBM Watson Studio, CSV , EJS, CSS, Bootstrap, Node.js, Express.js, MongoDB Atlas

PROJECTS

Preventive AI ↗

📅 Sept 2024 - Oct 2024



- An AI platform combining medical imaging, genomic data, and EHRs for early cancer detection and personalized treatment insights.
- Tech Stack: Python, Supervised ML, EJS, CSS, Node.js, Express.js, MongoDB Atlas, TensorFlow Serving

Saarthi ↗

📅 Nov 2024



- Built a full stack application offering free educational tools, including problem-solving resources, an NEP syllabus generator, AI chatbot, Doc QnA , Quiz, and more.
- Tech Stack: Bootstrap, Gemini API, Plotch AI, Node.js, Express.js, MongoDB Atlas, Vertex AI, Groclake

Hack-Match ↗

📅 Jan 2024



- Developed a platform to help users find ideal hackathon teammates by matching them based on skill sets & interests.
- Tech Stack: EJS, CSS, Cloudinary, Bootstrap, Render, Node.js, Express.js, NoSQL DB

User Authentication System ↗

📅 May 2024



- Built a full-stack, responsive user authentication website with secure registration, login, and profile management
- Tech Stack: EJS, CSS, Bootstrap, Render, Node.js, Express.js, NoSQL DB

SOFT SKILLS

Leadership, Decision Making, Teamwork, Creativity

EDUCATION

Bachelor of Technology, Computer Science

Cochin University Of Science And Technology (CUSAT)

📍 Kochi, Kerala

📅 Oct 2022 – May 2026 (Expected)

Currently in 4th Year - CGPA : 9.6

Academic Achievements

- Represented INDIA at European Rover Challenge, 2024, Poland - Robotics Competition. Team name : Horizon (World Rank 18th)
- Listed among Top 10% Students in the Batch.

POSITIONS OF RESPONSIBILITY

- Team Lead @HORIZON- Mars Rover Team, CUSAT
- SIG AI Lead @IEEE Execom'24- CS Branch Society,CUSAT

SKILLS

Languages

Java Python JS C/C++ Embedded C Ruby

Frameworks and Libraries

React OpenGL ROS NodeJS ExpressJS
RestAPI AXIOS Gemini API Pandas
AWS STM32CUBE - IDE LLMware

Databases

MongoDB SQL

Tools

Git/GitHub Linux Vertex AI OOP AI-ML Proteus

ACHIEVEMENTS

- Won 5 Hackathons (out of 10) conducted by Google Development Group ↗ (GDG) , Major League Hacking ↗ (MLH) & Inovact ↗.
- Solved 100+ programming problems across various platforms like LeetCode, Hackerrank etc.
- Jury Choice Award Winner at AIC by Accenture & Finalist at DevsHouse'24 Hackathon by VIT Chennai

CERTIFICATIONS

ERC Space & Robotics Competition, AGH University Poland ↗

📅 Sep 2024

Certificate of Appreciation (Tech. Team) - TEDxCUSAT↗

📅 Sep 2024

Artificial Intelligence Fundamentals, IBM ↗

📅 June 2024

Cyber Security and Ethical Hacking , IIT Delhi ↗

📅 Nov 2023

Shaastra X Codechef Programming Tournament - IIT Madras ↗

📅 Oct 2023